

through removable storage drives like a USB flash drive.

Elk Cloner, Cascade, Brain, Ghostball and Ping-Pong are examples of computer viruses. Elk Cloner was the first computer virus detected in 1982. It was able to attach itself to the Apple II file system and was spread through floppy disks. With antivirus software becoming more powerful, computer infections are caused more often by worms than by viruses these days. That's because viruses require a modification of the files in order to replicate but worms don't, and powerful antivirus software prevents unauthorised access to protected files.

- **Worms:** As the name suggests, a computer worm program can jump just like a worm from one location to another, but the location can only be a host computer. When a worm invades a computer, it doesn't infect any file like a virus, but spreads by copying itself from one computer to another through a connected network. After entering a host computer, it starts scanning the network to find any other connected computers (hosts) and when it does so, it transfers itself to them using a recursive method. The main purpose of a worm is to control the host computer or a group of computers connected to a network. Unlike viruses, worms don't always harm the files stored in the computer but can harm the network by using the bandwidth for controlling the computers.

Morris, Mydoom, Here you have, Shamoon and Linux.Darloz are examples of computer worms. Among these, Morris was the first one to be detected in 1988. Written by Robert Tappan Morris, who was a graduate student at Cornell University, it was launched on November 2, 1988, from computer systems at the Massachusetts Institute of Technology. This led to the first felony conviction in the US under the 1986 Computer Fraud and Abuse

Act. Mydoom was another very bad worm attack and caused an estimated damage of US\$ 38 billion in 2004.

This worm spread through mass emailing. It first infected a computer that opened the infected emails. It then extracted email addresses from this computer and sent emails to them. It was also able to connect other infected computers over the Internet, forming a web of computers called a botnet, which was able to cause Distributed Denial of Service (DDoS) attacks.

- **Trojan horse:** A Trojan horse is a kind of malware that can deceive the user. That's why the term is derived from the ancient Greek story of a deceptive Trojan horse that led to the fall of the city of Troy. This type of malware can come with other software as a package, and the user runs the software thinking it is reliable. That's why most pirated software or games come with such Trojan horse malware, and you can install these without knowing about the malware injected into the files. However, most Trojan horses are spread through social media sites and emails. The attacker sends an email containing a link or attachment with the infected files and the user is duped to click on the link or run the file – which is also known as phishing. That's why it's a good practice not to click on any links or open file attachments from unknown senders. Once the Trojan horse is executed by the user unknowingly, it starts running in the background and executes the payload. The payload can be anything, but most often is capable of controlling the user's computer and stealing information. Trojans may allow an attacker to access a user's personal information such as banking information, passwords or personal identity, and can also delete a user's files or infect other devices connected to the network.

EGABTR, DarkComet, Blackhole exploit kit, Gh0st RAT and Kedi RAT are some examples of Trojan horses. Among these, EGABTR was detected in the early 1980s. It was actually a graphics utility to improve the EGA display but the Trojan payload would delete the File Allocation Table (FAT) of the hard drive by displaying a message "Arf! Arf! Got you!". Kedi RAT was a Trojan detected in 2017 and belongs to the Remote Access Trojan (RAT) category. RAT malware can infect a computer and give remote access to a hacker, which means the latter is able to control and access the infected computer from another computer using the connected network. Kedi RAT was distributed in a Spear Phishing Campaign and it targeted Citrix users. This malware was able to evade many malware scanners, and was able to communicate with the control centre (hacker) via Gmail using HTML and HTTP protocols.

- **Ransomware:** This type of malware causes temporary denial of access to data and threatens the user by demanding a ransom amount. Some ransomware uses cryptoviral extortion techniques to encrypt the user's files, making it very difficult to access them, and informs the user that it will decrypt the files only after the ransom is paid. The payment is demanded through a cryptocurrency like Bitcoin so that the source of attack is not traceable. Most ransomware is delivered to a computer using a Trojan horse by duping the user to download an infected file or click on a link that activates a Trojan script. However, a worm can also deliver ransomware. For example, the WannaCry worm travelled automatically between computers without user interaction and worked as ransomware. AIDS, CryptoLocker, Locky, WannaCry and Thanatos are examples of ransomware. AIDS was the first ransomware detected in 1989 and was